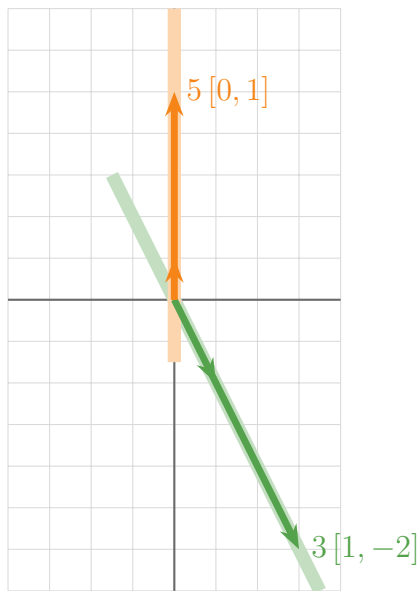


Eigenvectors of A

$$\lambda = 3 \text{ and } 5$$



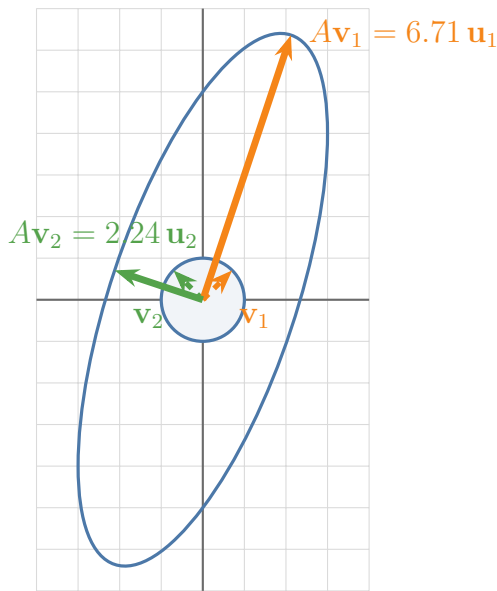
each keeps its own line,
but the two lines are not perpendicular

dashed: before A

solid: after A

Singular vectors of A

$$\sigma_1 = 6.71, \sigma_2 = 2.24$$



the unit circle becomes an ellipse;

perpendicular $\mathbf{v}_1, \mathbf{v}_2$ land on its perpendicular axes

$$A = \begin{bmatrix} 3 & 0 \\ 4 & 5 \end{bmatrix}$$